



THE  
KING'S  
SCHOOL

2019

Year 11  
Assessment Task 3 (AT3)  
September 2019

# Mathematics Extension 1

## General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11-14, show relevant mathematical reasoning and/or calculations
- Write your **NAME** where indicated

Total marks: 70

## Section I 10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

## Section II 60 marks

- Attempt Questions 11-14
- Allow about 1 hour and 45 minutes for this section

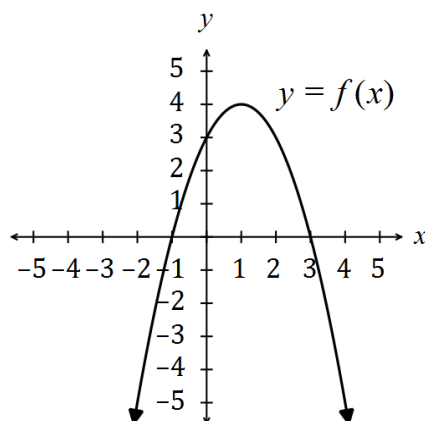
| Examiners' Use Only |                 |                  | Setter/Checker: External/BDB |       |            |
|---------------------|-----------------|------------------|------------------------------|-------|------------|
| Question            | Calculus        | Functions        | Trigonometric Functions      | Total |            |
| <b>Section I</b>    |                 |                  |                              |       |            |
| <b>1-10</b>         | 8 / 1           | 1, 3-5, 7, 9 / 6 | 2, 6, 10 / 3                 |       | <b>/10</b> |
| <b>Section II</b>   |                 |                  |                              |       |            |
| <b>11</b>           |                 | <i>a-c</i> /12   | <i>d</i> / 3                 |       | <b>/15</b> |
| <b>12</b>           | <i>b</i> / 7    | <i>e</i> / 4     | <i>a, c, d</i> / 4           |       | <b>/15</b> |
| <b>13</b>           | <i>c, d</i> / 7 | <i>a</i> / 4     | <i>b</i> / 4                 |       | <b>/15</b> |
| <b>14</b>           | <i>e</i> / 5    | <i>c, d</i> / 6  | <i>a, b</i> / 4              |       | <b>/15</b> |
| <b>Total</b>        | /20             | /32              | /18                          |       | <b>/70</b> |

**Section I**  
**10 marks**

Attempt Questions 1-10  
Allow about 15 minutes for this section

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- 1 Consider the function  $y = f(x)$  below:



What is the domain of  $y = \frac{1}{f(x)}$ ?

- (A)  $(-\infty, 0) \cup \left[\frac{1}{4}, \infty\right)$   
(B)  $(-\infty, 0) \cup \left(\frac{1}{4}, \infty\right)$   
(C)  $[-\infty, -1) \cup (-1, 3) \cup (3, \infty]$   
(D)  $(-\infty, -1) \cup (-1, 3) \cup (3, \infty)$

- 2 Simplify  $4\sin 2\theta\cos 2\theta$

- (A)  $\sin \theta$   
(B)  $2\sin \theta$   
(C)  $\sin 4\theta$   
(D)  $2\sin 4\theta$

3 Solve  $|3 - 2x| \leq 1$

(A)  $1 \leq x \leq 2$

(B)  $x \leq 1$  or  $x \geq 2$

(C)  $-2 \leq x \leq -1$

(D)  $x \leq -2$  or  $x \geq -1$

4 Find the inverse relation of  $y = x^2 - 4x + 17$

(A) No real solution

(B)  $y = 2x - 4$

(C)  $y = 4 \pm \sqrt{x - 1}$

(D)  $y = 2 \pm \sqrt{x - 13}$

5 If  $\alpha, \beta, \gamma$  and  $\delta$  are the roots of  $x^4 - x^3 + 4x^2 - 2x + 4 = 0$ , evaluate  $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$

(A)  $-\frac{1}{2}$

(B)  $\frac{1}{2}$

(C) 1

(D) -1

6 Solve:  $\sin^{-1} x + 2\sin x \cos x + \cos^{-1} x = \frac{1 + \pi}{2}, 0 \leq x \leq \frac{\pi}{4}$

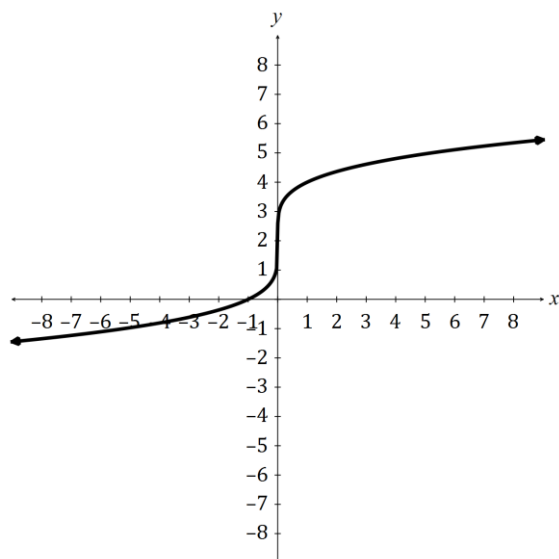
(A)  $\frac{\pi}{12}$

(B)  $\frac{\pi}{6}$

(C)  $\frac{\pi}{4}$

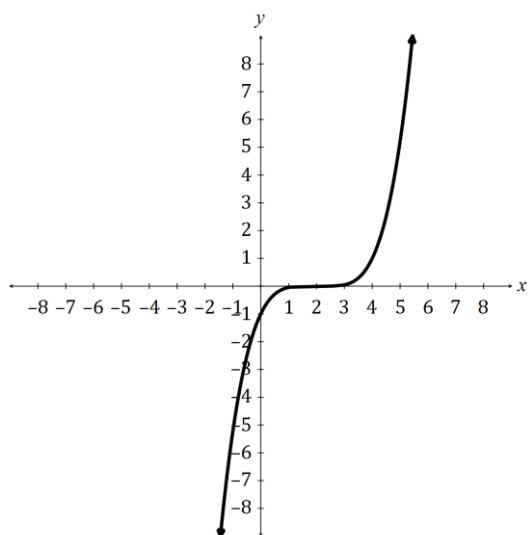
(D) No real solution

7 Consider the graph  $y = f(x)$  below.

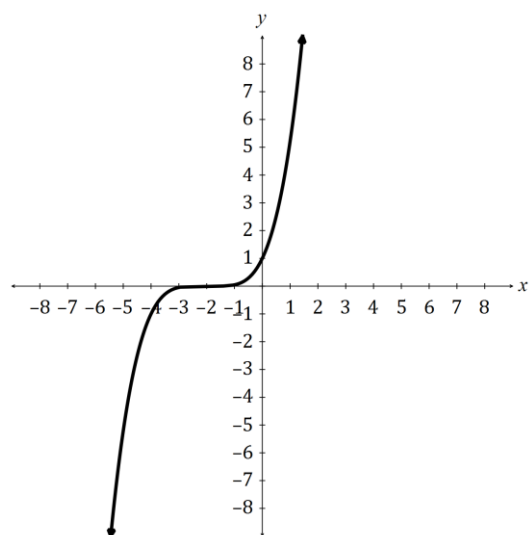


Which graph represents  $y = f^{-1}(x)$ ?

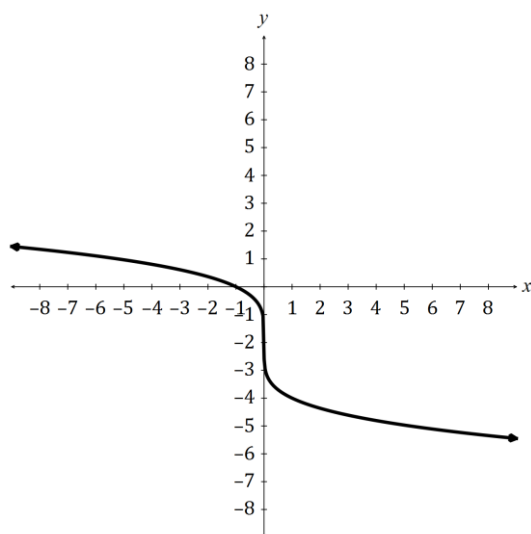
(A)



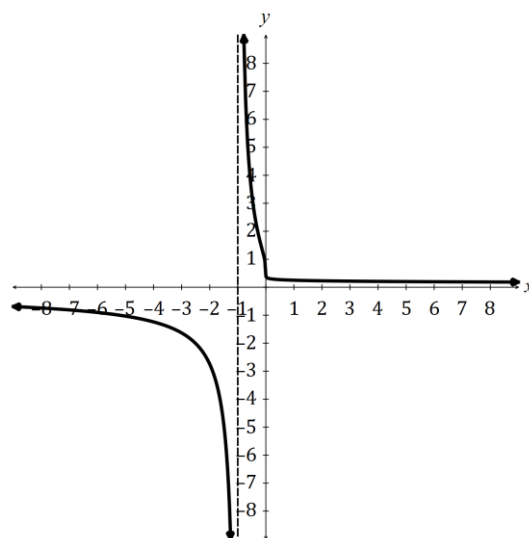
(B)



(C)



(D)



- 8 The velocity of a particle which is travelling along the  $x$ -axis is given by

$$v = 3t^2 + 4t + 5, \text{ where } t \text{ is the time in seconds.}$$

Which of the following could be the function which gives the position of the particle with respect to time?

(A)  $x = t^3 + 2t^2 + 5t + 10$

(B)  $x = 3t^3 + 4t^2 + 5t + 10$

(C)  $x = 6t + 4$

(D)  $x = 3t + 4$

- 9 If  $P(x) = 0$  has a root of multiplicity 3 at  $x = k$ , then  $P'(k) = 0$  has

(A) a root of multiplicity  $\leq 3$

(B) a root of multiplicity 3

(C) a root of multiplicity 2

(D) a rational root of multiplicity  $\leq 2$

- 10 If  $\sin\theta = \sqrt{\frac{x}{3}}$  ( $x \geq 0$ ), find the exact value of  $\cos 2\theta$

(A)  $\frac{3 - 2x}{3}$

(B)  $\frac{2\sqrt{3x - x^2}}{3}$

(C)  $\frac{-2\sqrt{3x - x^2}}{3}$

(D)  $\sqrt{\frac{3 - 2x}{3}}$

**End of Section I**

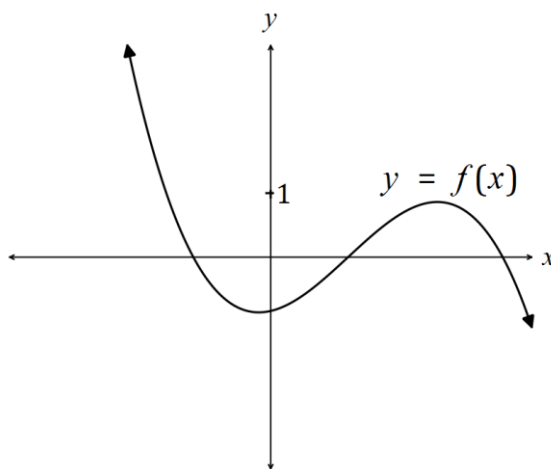
**Section II**  
**60 marks**

Attempt Questions 11-14  
Allow about 1 hour 45 minutes for this section

**Question 11** (15 marks)

**Marks**

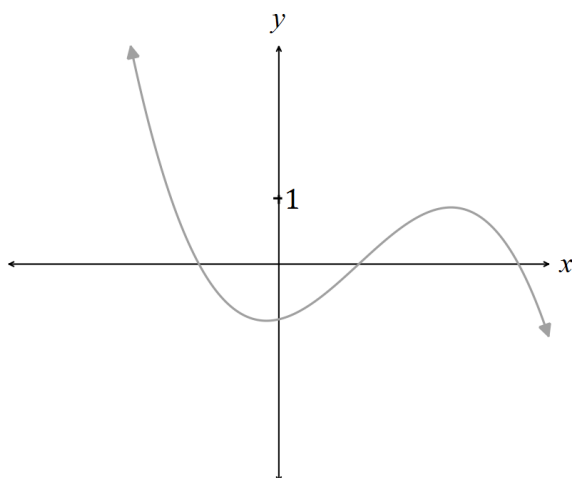
(a) Consider the function  $y = f(x)$  below:



On each number plane provided, sketch the function indicated:

(i)  $y = |f(x)|$

**1**

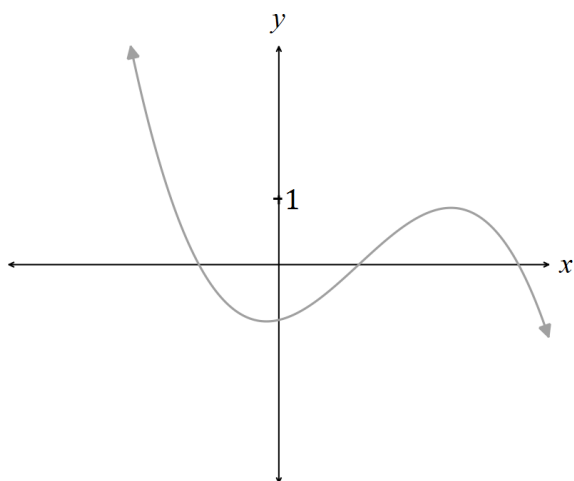


**Question 11 (continued)**

**Marks**

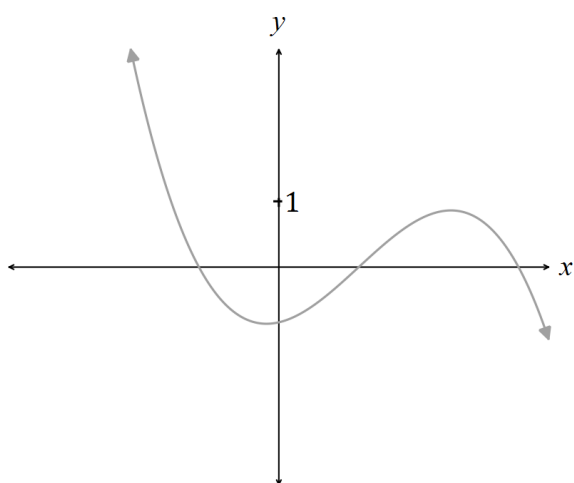
(a) (ii)  $y = f(|x|)$

**1**



(iii)  $y^2 = f(x)$

**1**



**Question 11 continues on the next page**

**Question 11 (continued)**

**Marks**

(b) Solve:

(i)  $x^2 < 6 - x$

**2**

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(ii)  $\frac{x + 3}{1 - x} \geq 1$

**3**

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**Question 11 continues on the next page**

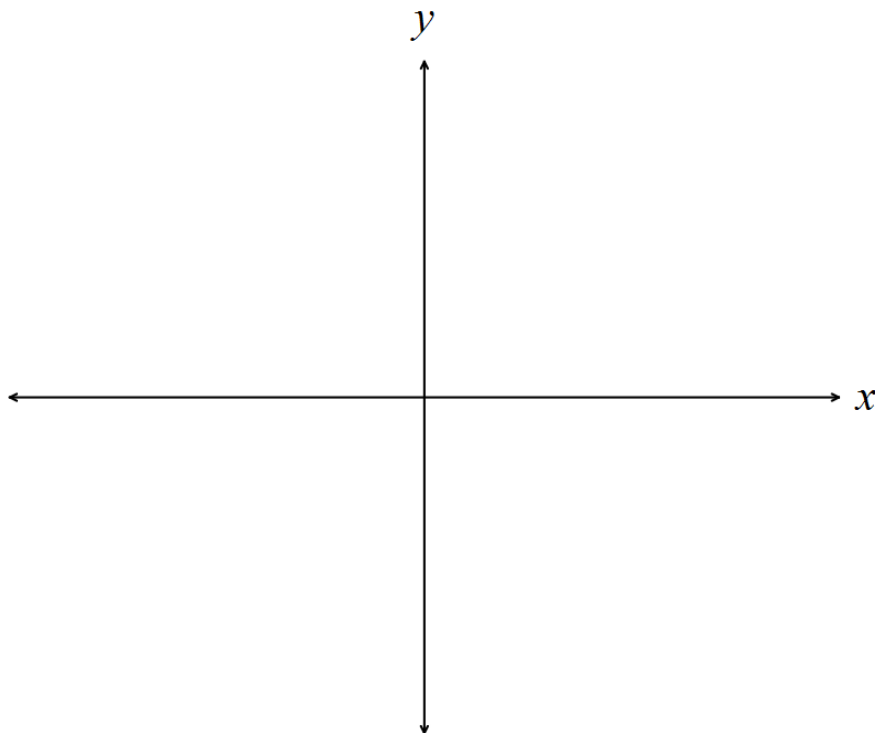
**Question 11 (continued)**

**Marks**

(c) For the polynomial function  $P(x) = x^2 (2 - x)^3 (3 + x)$

(i) Sketch  $y = P(x)$  on the number plane provided.

**2**



(ii) Hence, solve  $P(x) \leq 0$

**2**

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**Question 11 continues on the next page**

**Question 11 (continued)**

**Marks**

(d) Find the exact value of  $\sin 15^\circ$

**3**

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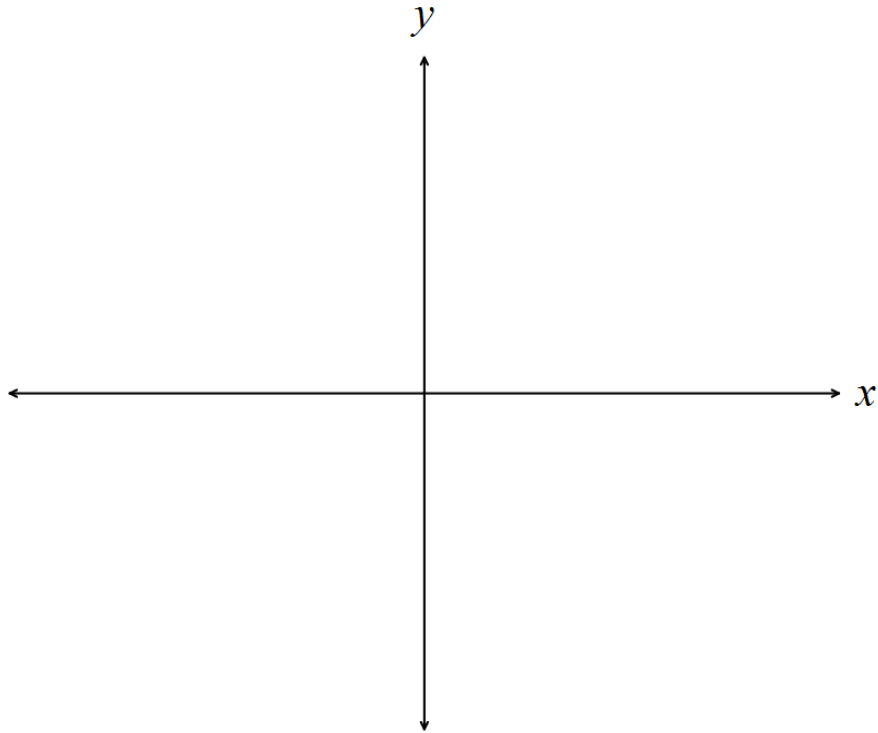
**End of Question 11**

**Question 12** (15 marks)

**Marks**

(a) Sketch the graph  $y = \cos^{-1} x$

**2**



**Question 12 continues on the next page**

**Question 12** (continued)

**Marks**

- (b) The displacement of a particle moving along the  $x$ -axis is given by  $x = (t - 1)^3$ , where  $t$  is time in seconds and  $x$  is displacement in cm.

- (i) Find the equations for velocity ( $\dot{x}$ ) and acceleration ( $\ddot{x}$ ). **2**

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- (ii) Find the acceleration of the particle after 10 seconds. **1**

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- (iii) When is the particle at rest? **1**

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- (iv) What is the minimum acceleration of the particle? **1**

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- (v) Describe the motion of the particle. **2**

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**Question 12 continues on the next page**

**Question 12** (continued)

**Marks**

(c) Simplify  $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$

**1**

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(d) For what domain does  $\tan (\tan^{-1} x) = x$ ?

**1**

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**Question 12 continues on the next page**

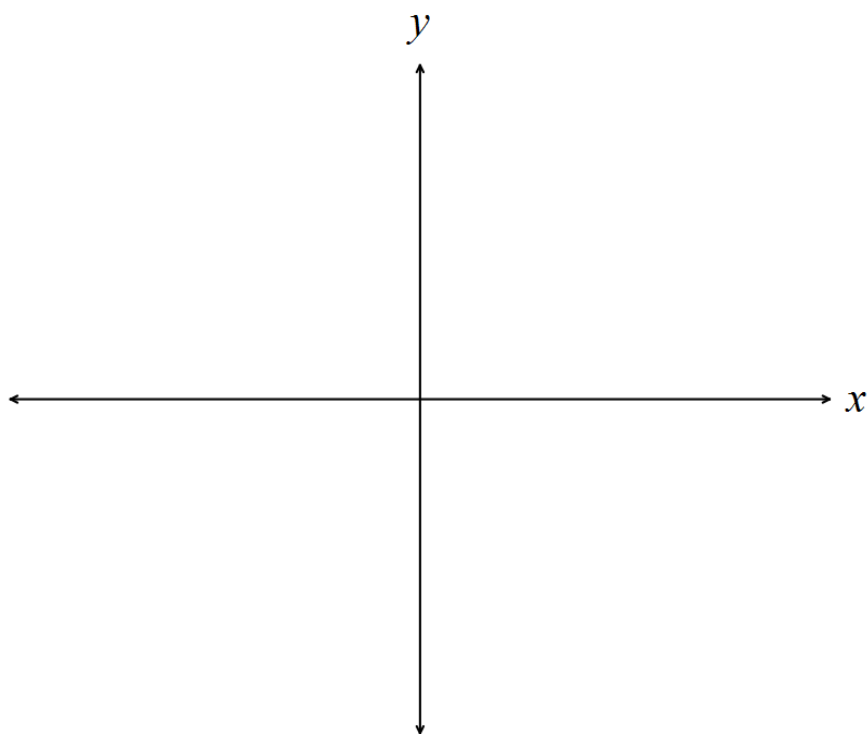
**Question 12** (continued)

**Marks**

(e) Consider the Cartesian equation  $x^2 + y^2 - 10y - 11 = 0$

(i) Sketch the graph.

**2**



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(ii) Given  $x = 6 \sin \theta$ , find  $y$  in terms of  $\theta$ .

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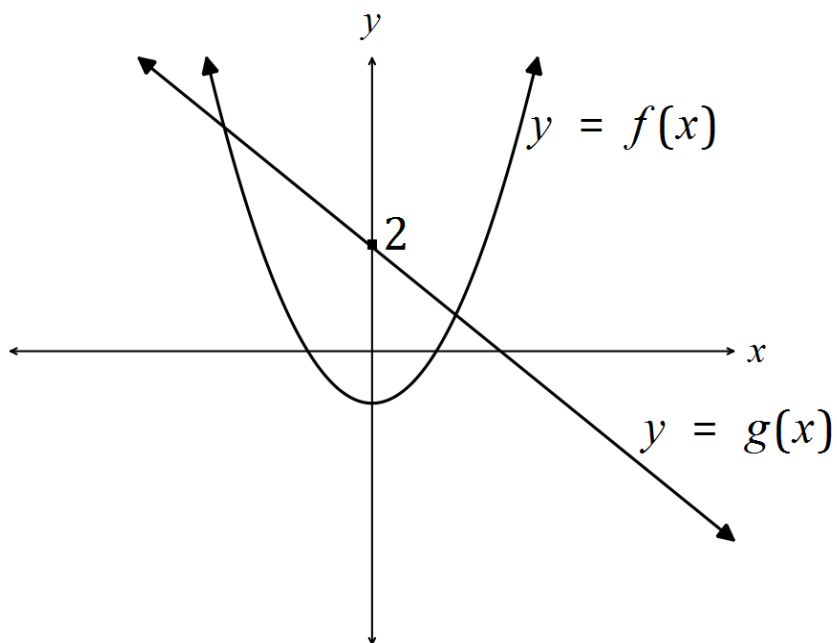
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**End of Question 12**

**Question 13** (15 marks)

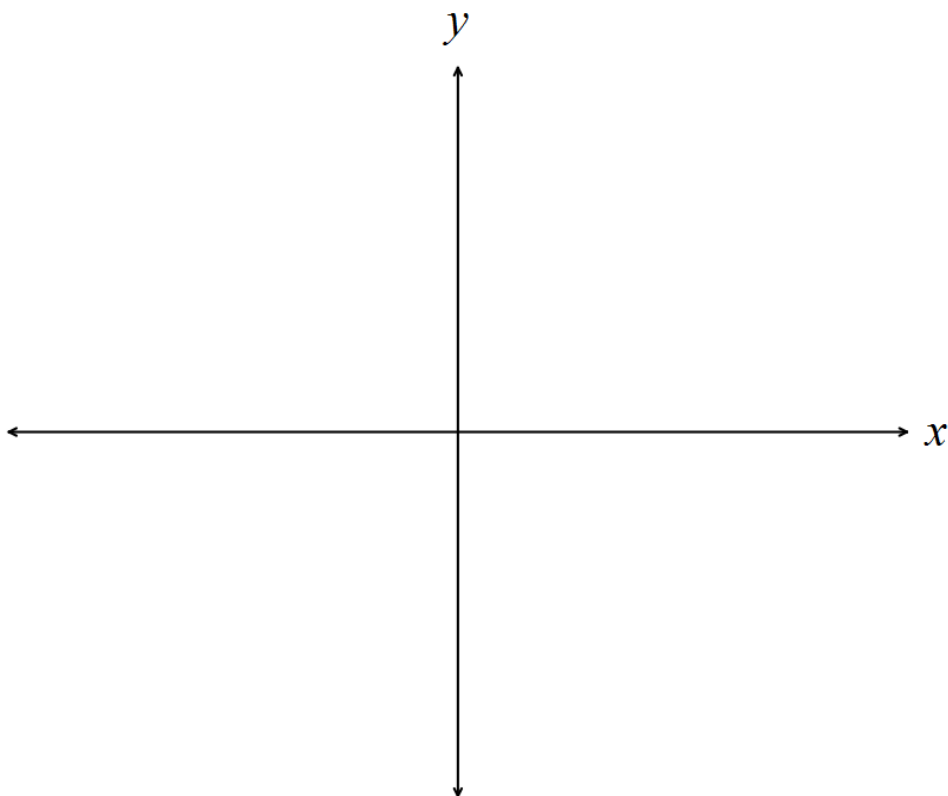
**Marks**

(a) The graphs of  $y = f(x)$  and  $y = g(x)$  are shown:



(i) Sketch  $y = f(x) + g(x)$

2



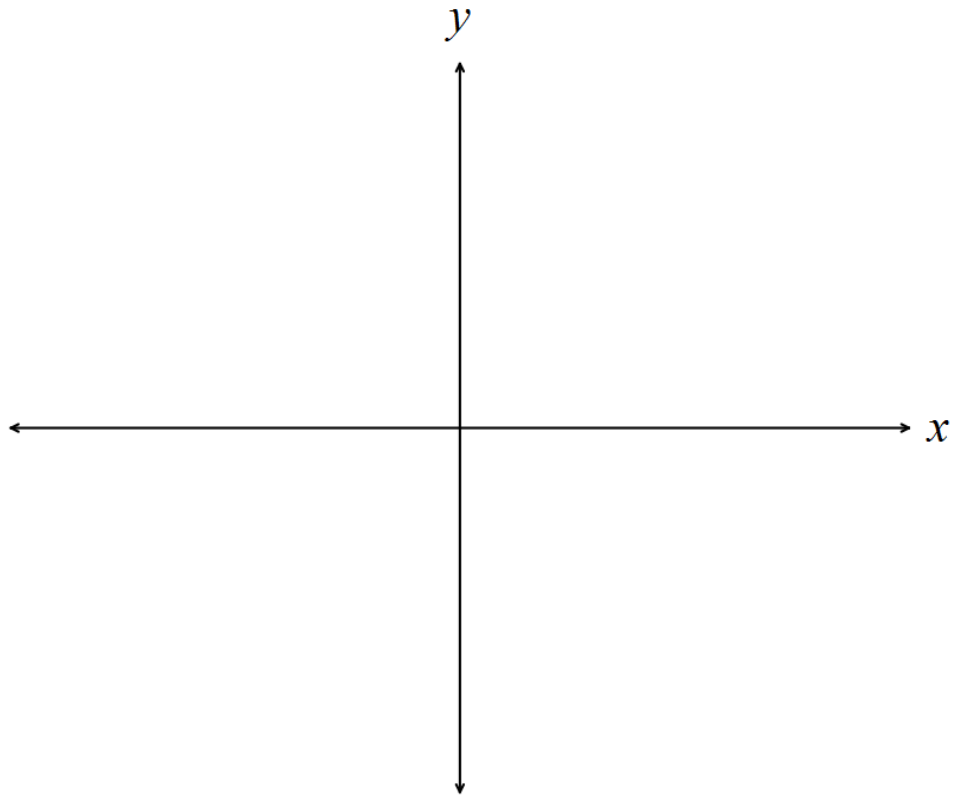
**Question 13 continues on the next page**

**Question 13** (continued)

**Marks**

(a) (ii) Sketch  $y = f(x) \cdot g(x)$

**2**



**Question 13 continues on the next page**

**Question 13** (continued)

**Marks**

(b) (i) Prove  $\cos(A + B) + \cos(A - B) \equiv 2\cos A \cos B$

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(ii) Hence, find the exact value of  $\cos \frac{5\pi}{12} + \cos \frac{\pi}{12}$

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**Question 13 continues on the next page**

**Question 13** (continued)

**Marks**

(c) The number of bacteria in a culture is given by  $N = N_0 e^{0.28t}$  where  $t$  is in hours.

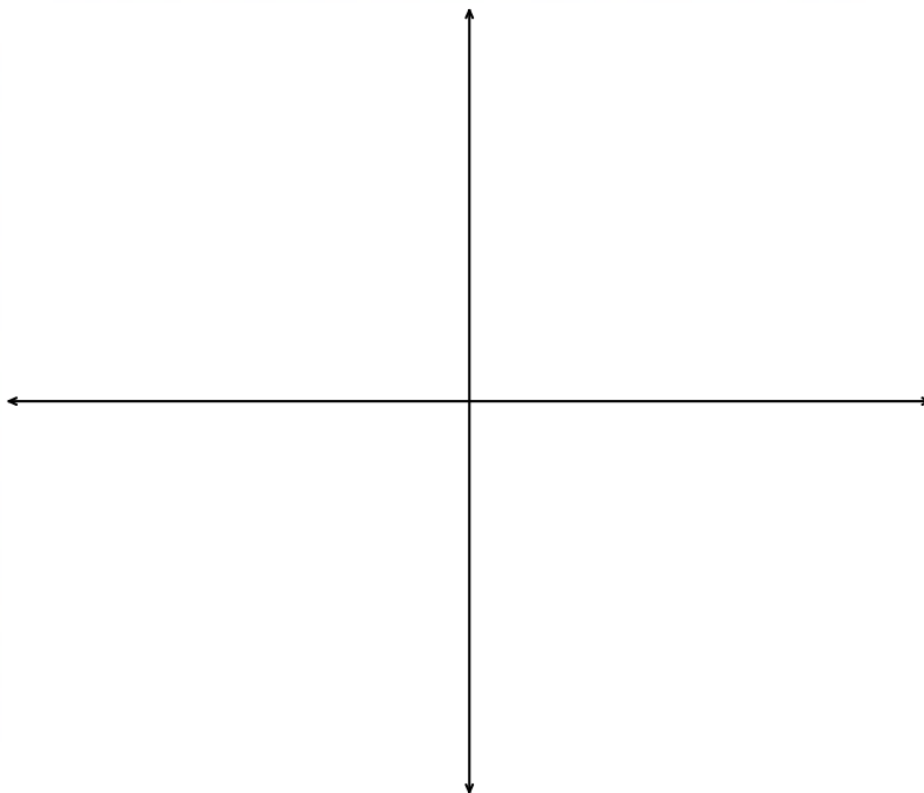
(i) If there are initially 100 bacteria, how many will there be after 10 hours? **1**

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(ii) How many hours will it take for the population to reach 200 000? Answer to the nearest hour. **2**

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(iii) Graph the relationship between  $N$  and  $t$  for  $N \in [100, 200\,000]$  **1**



**Question 13 continues on the next page**

**Question 13** (continued)

**Marks**

- (d) A spherical balloon has a radius  $r$  mm. It has a small hole, causing the radius to shrink at a rate of  $0.2 \text{ mm s}^{-1}$ .

Calculate the rate of change of the surface area of the balloon when the radius is 100 mm.

Hence, describe what is happening to the surface area of the balloon.

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**End of Question 13**

**Question 14** (15 marks)

**Marks**

(a) Prove  $\sin^{-1}(-x) = -\sin^{-1}x$

**2**

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(b) Show that  $\tan(\sin^{-1}x) = \frac{x}{\sqrt{1-x^2}}$ , for  $0 \leq x < 1$

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**Question 14 continues on the next page**

**Question 14** (continued)

**Marks**

(c) Given  $P(x) = 6x^3 - 11x^2 - 3x + 2$

(i) Show  $x - 2$  is a factor of  $P(x)$

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(ii) Hence, factorise  $P(x)$  and find the roots of  $P(x) = 0$

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**Question 14 continues on the next page**

**Question 14** (continued)

**Marks**

- (d) Consider the polynomial  $P(x) = (x - 2)(x - 5)Q(x) + ax + b$ , where  $Q(x)$  is a polynomial and  $a$  and  $b$  are real numbers.

When  $P(x)$  is divided by  $(x - 2)$  the remainder is 4, and when it is divided by  $(x - 5)$ , the remainder is 13.

Find  $a$  and  $b$

**2**

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**Question 14 continues on the next page**

**Question 14** (continued)

**Marks**

- (e) The blood temperature of a live animal is  $38.6^{\circ}\text{C}$ .

An animal is found dead with a blood temperature of  $31.1^{\circ}\text{C}$ , in a barn of constant temperature  $P^{\circ}\text{C}$ . 10 minutes later, the animal's blood temperature is  $30.7^{\circ}\text{C}$ .

The cooling rate is proportional to the difference between the blood temperature and the room temperature.

- (i) By using the formula for Newton's Law of cooling,  $T = P + Ae^{kt}$ , find an expression for  $k$  in terms of  $A$ .

**2**

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- (ii) Given  $P = 26^{\circ}$ , show that  $k = -0.008$  to three decimal places.

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**Question 14 continues on the next page**

**Question 14** (continued)

**Marks**

- (e) (iii) Hence, if the animal was found at 8 pm, what was the time of death? Answer to the nearest minute.

**2**

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**End of Examination Paper**



THE  
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2019

Year 11  
Assessment Task 3 (11AT3)  
September 2019

Multiple Choice Answer Sheet  
Mathematics Extension 1

Select the alternative A, B, C or D that best answers the question and indicate your choice by filling in the appropriate response oval, completely.

|    |     |     |     |     | Examiner's Use Only     |     |
|----|-----|-----|-----|-----|-------------------------|-----|
| 1  | (A) | (B) | (C) | (D) | Functions               | / 1 |
| 2  | (A) | (B) | (C) | (D) | Trigonometric Functions | / 1 |
| 3  | (A) | (B) | (C) | (D) |                         |     |
| 4  | (A) | (B) | (C) | (D) |                         |     |
| 5  | (A) | (B) | (C) | (D) | Functions               | / 3 |
| 6  | (A) | (B) | (C) | (D) | Trigonometric Functions | / 1 |
| 7  | (A) | (B) | (C) | (D) | Functions               | / 1 |
| 8  | (A) | (B) | (C) | (D) | Calculus                | / 1 |
| 9  | (A) | (B) | (C) | (D) | Functions               | / 1 |
| 10 | (A) | (B) | (C) | (D) | Trigonometric Functions | / 1 |
|    |     |     |     |     | Total                   | /10 |

| Examiners' Use Only |          |                  |                         | Setter/Checker: External/BDB |  |
|---------------------|----------|------------------|-------------------------|------------------------------|--|
| Question            | Calculus | Functions        | Trigonometric Functions | Total                        |  |
| Section I           |          |                  |                         |                              |  |
| 1-10                | 8 / 1    | 1, 3-5, 7, 9 / 6 | 2, 6, 10 / 3            | /10                          |  |
| Total               | /20      | /32              | /18                     | /70                          |  |

**2019 Year 11 Examination  
Mathematics Extension 1  
Marking Guidelines**

**Section I**

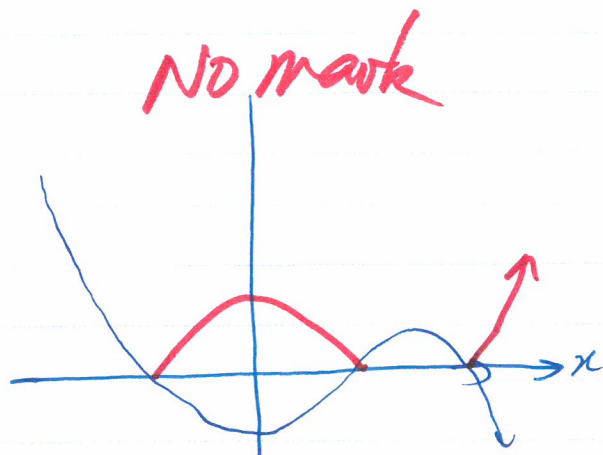
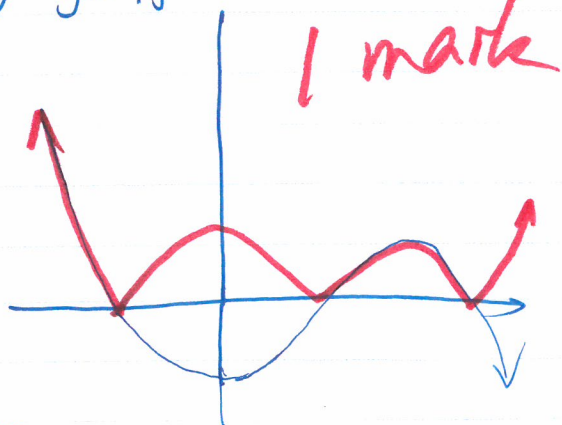
AT3

**Multiple-choice Answer Key**

| Question | Answer |
|----------|--------|
| 1        | D      |
| 2        | D      |
| 3        | A      |
| 4        | D      |
| 5        | B      |
| 6        | A      |
| 7        | A      |
| 8        | A      |
| 9        | C      |
| 10       | A      |

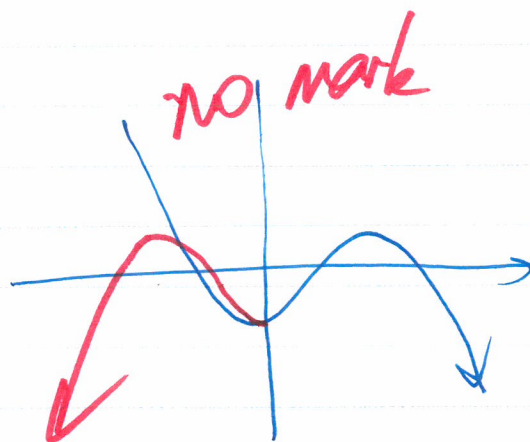
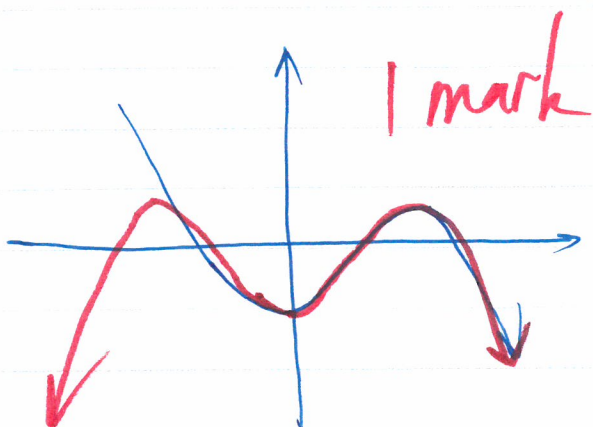
# Year 11 AT3 Ex) Question 11 MARKING.

(a) i)  $y = |f(x)|$

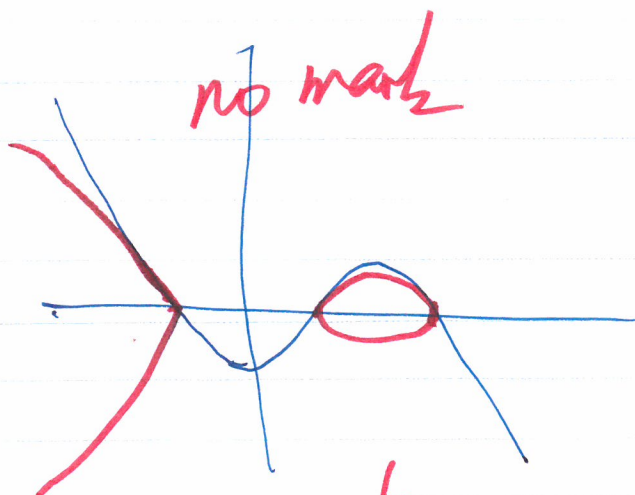
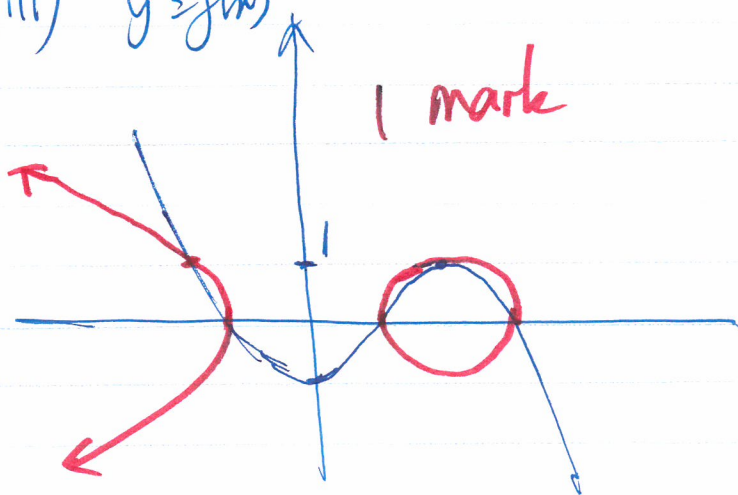


(ii)

$y = f(|x|)$



(iii)  $y = f(x)$

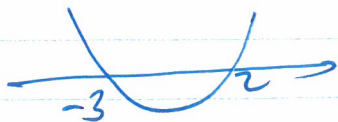


other variations  
= no mark

(b) (i)

$$x^2 + x - 6 < 0$$
$$(x+3)(x-2) < 0$$

← 1 for factored quad.



$$-3 < x < 2 \quad \text{or} \quad x \in (-3, 2).$$

← 1 correct answer

2

(ii)

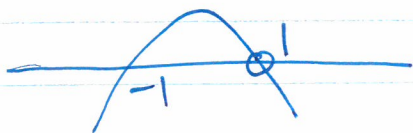
$$\frac{x+3}{1-x} \geq 1$$

$$(x+3)(1-x) \geq (1-x)^2$$

$$(x+3)(1-x) - (1-x)^2 \geq 0$$

$$2(1-x)(x+1) \geq 0.$$

1 mark for progress

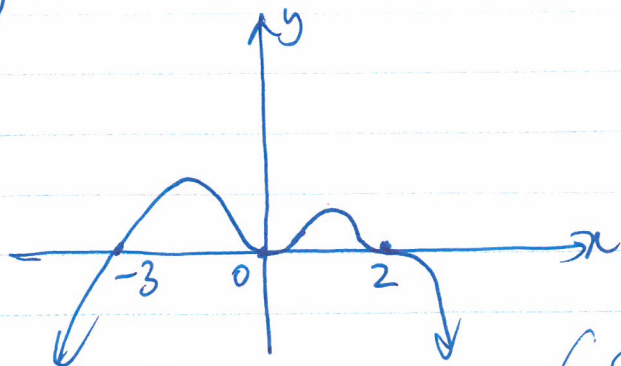


$$-1 \leq x \leq 1$$

← 2 marks correct answer  
(-1 for  $-1 \leq x \leq 1$ )  
mark 3

{ "Critical points method" correctly solve the problem,  
showing working to  $-1 \leq x < 1$  3  
 $-1 \leq x \leq 1$  2 }  
~~completely wrong~~ Very few used this method.

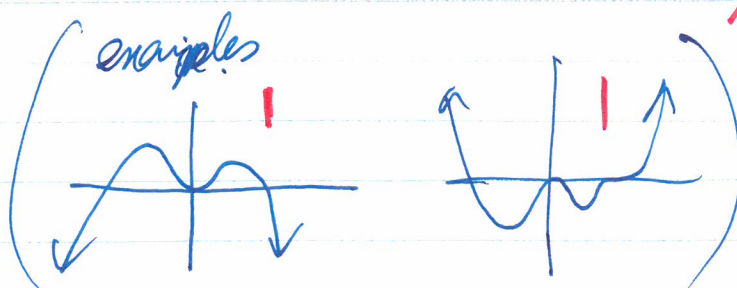
(c) (i)



Correct x intercepts  
and shape at intercepts 1

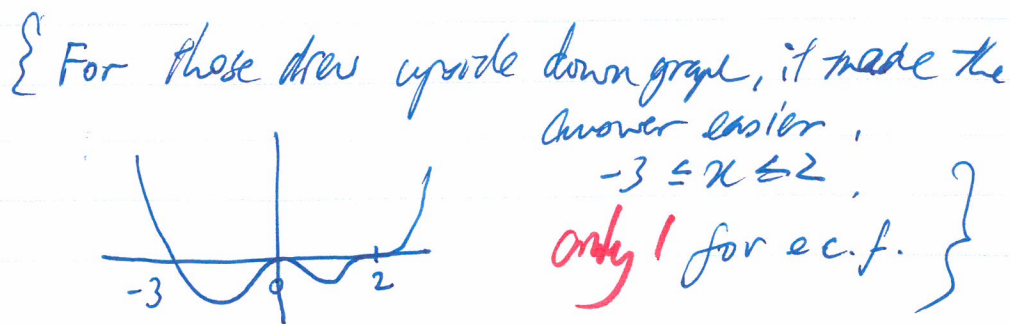
Correct <sup>even</sup> degree graph

1 1/2



(c) (ii)  $x \leq -3, x = 0, x \geq 2$  /2  
 -1 mark for ~~side~~ mistake

example  $x \leq -3, x \geq 2$  only 1 mark



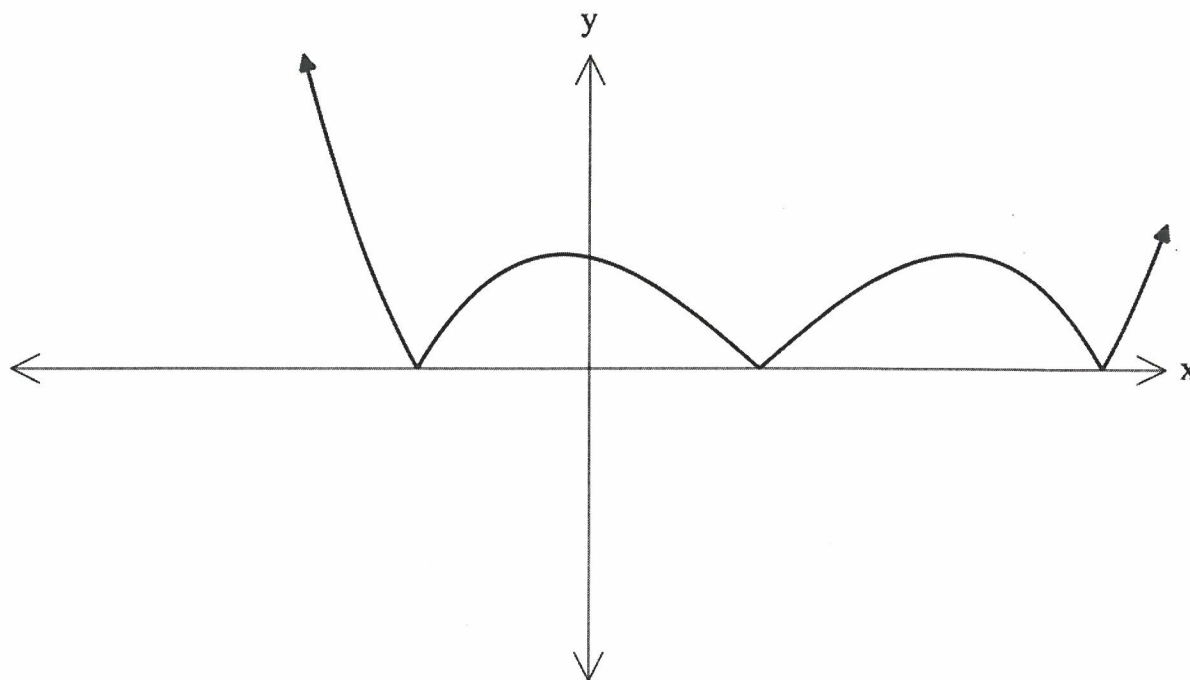
(d)  $\sin(15^\circ) = \sin(45^\circ - 30^\circ)$   
 $= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$  ← 1 mark for correct use of  $\sin(A-B)$ .  
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$   
 $= \frac{\sqrt{3}-1}{2\sqrt{2}}$  ← 2 mark with working shown. /3  
 (or  $\frac{\sqrt{6}-\sqrt{2}}{4}$ )  
 -1 mark for error.

{ 0 mark for  $\sin 15^\circ = \sin 45^\circ - \sin 30^\circ$  }

**Question 11**

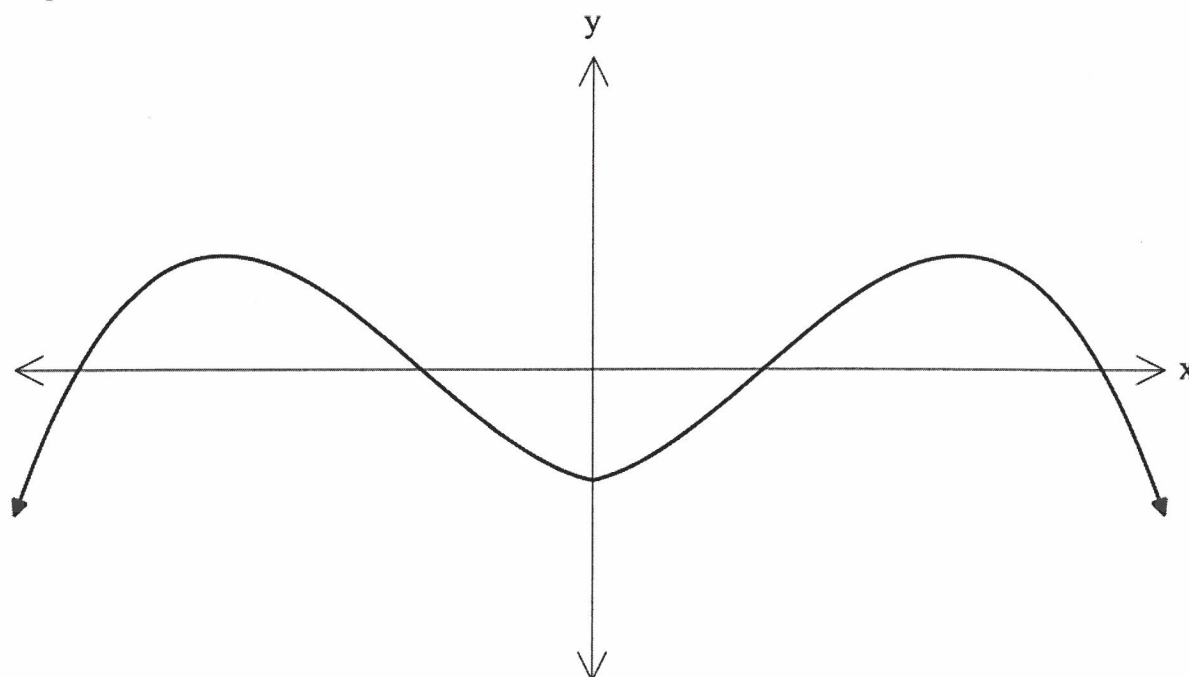
11 (a) (i)

| Criteria                 | Marks |
|--------------------------|-------|
| • Provides correct graph | 1     |

**Sample answer**

11 (a) (ii)

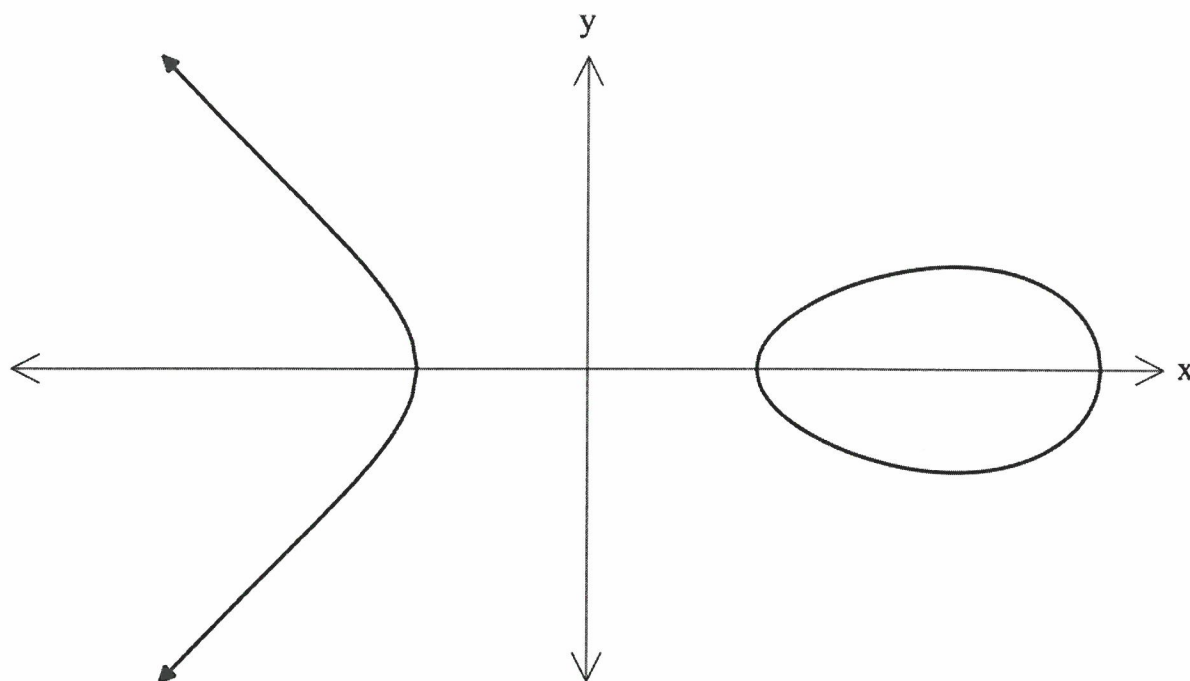
| Criteria                 | Marks |
|--------------------------|-------|
| • Provides correct graph | 1     |

**Sample answer**

11 (a) (iii)

| Criteria                 | Marks |
|--------------------------|-------|
| • Provides correct graph | 1     |

**Sample answer**



**Question 11 (continued)**

**Marks**

(b) Solve:

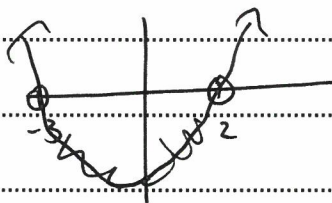
(i)  $x^2 < 6 - x$

2

$$x^2 + x - 6 < 0$$

$$(x+3)(x-2) < 0$$

①



$$\therefore -3 < x < 2$$

①

(ii)  $\frac{x+3}{1-x} \geq 1$

3

$$(x+3)(1-x) \geq (1-x)^2$$

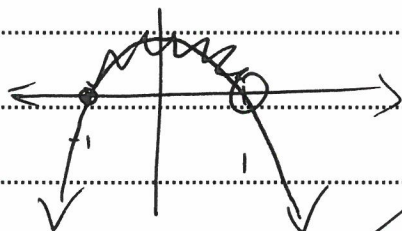
$$(x+3)(1-x) - (1-x)^2 \geq 0$$

$$(1-x)(x+3-1+x) \geq 0$$

$$(1-x)(2x+2) \geq 0$$

$$x \neq 1$$

①



$$-1 \leq x < 1$$

①

**Question 11 continues on the next page**

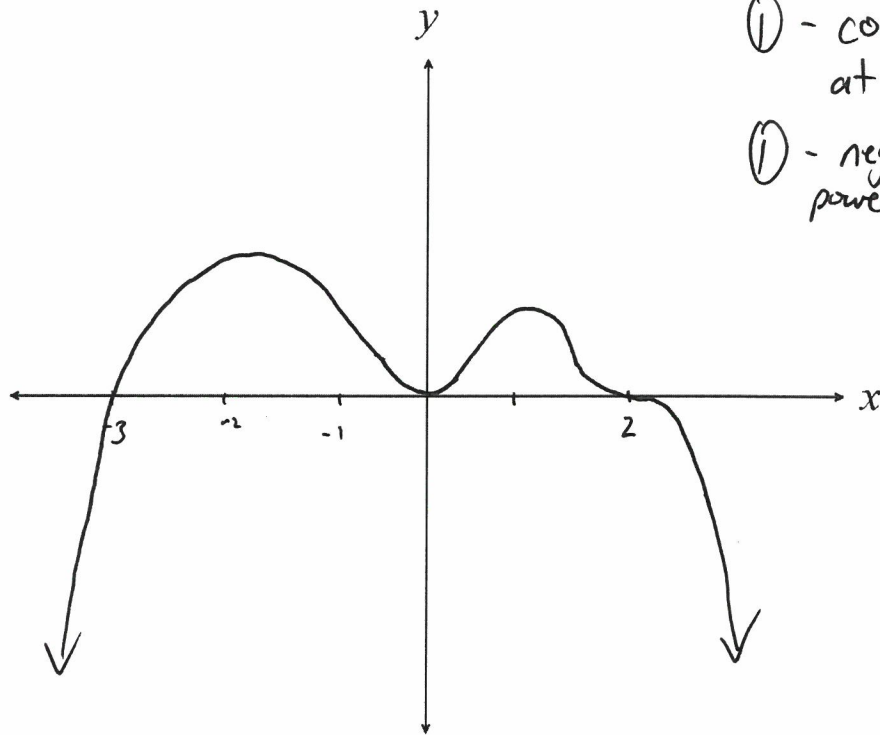
**Question 11 (continued)**

**Marks**

(c) For the polynomial function  $P(x) = x^2(2 - x)^3(3 + x)$

(i) Sketch  $y = P(x)$  on the number plane provided.

2



① - correct shape at roots.

① - negative even powered function.

(ii) Hence, solve  $P(x) \leq 0$

2

$$x \leq -3, x = 0, x \geq 2$$

(-1 mark for each incorrect.)

**Question 11 continues on the next page**

11 (d)

| Criteria                                          | Marks |
|---------------------------------------------------|-------|
| • Correct answer                                  | 3     |
| • Significant progress towards solution (1 error) | 2     |
| • Correctly applies the Sine difference formula   | 1     |

**Sample answer**

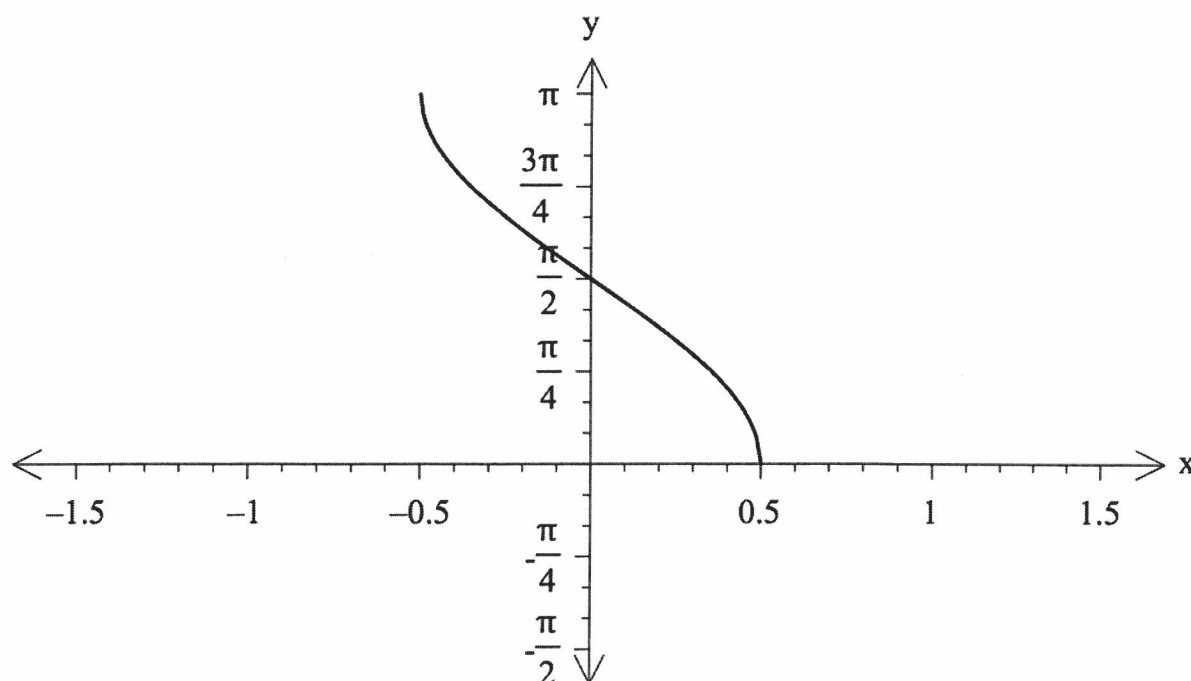
$$\begin{aligned}
 \sin 15^\circ &= \sin (45 - 30) \\
 &= \sin 45 \cdot \cos 30 - \cos 45 \cdot \sin 30 \\
 &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\
 &= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} \\
 &= \frac{\sqrt{3} - 1}{2\sqrt{2}} \\
 &= \frac{\sqrt{6} - \sqrt{2}}{4}
 \end{aligned}$$

**Question 12**

12 (a)

| Criteria                                                                      | Marks |
|-------------------------------------------------------------------------------|-------|
| • Correct solution                                                            | 2     |
| • Correct shape but incorrect range and/or domain, or intercepts not labelled | 1     |

**Sample answer**



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12 (b) (i)

| Criteria                 | Marks |
|--------------------------|-------|
| • Both equations correct | 2     |
| • One equation correct   | 1     |

**Sample answer**

$$\begin{aligned}
 x &= (t - 1)^3 \\
 \dot{x} &= 3(t - 1)^2 \\
 \ddot{x} &= 6(t - 1)
 \end{aligned}$$

12 (b) (ii)

| Criteria                                | Marks |
|-----------------------------------------|-------|
| • Correct solution, with correct units. | 1     |

**Sample answer**

at  $t = 10$

$$\begin{aligned}
 \ddot{x} &= 6(10 - 1) \\
 &= 54 \text{ cm s}^{-2}
 \end{aligned}$$

12 (b) (iii)

| Criteria           | Marks |
|--------------------|-------|
| • Correct solution | 1     |

**Sample answer**

for stationary points,  $\dot{x} = 0$

$$\therefore t = 1 \text{ s}$$

12 (b) (iv)

| Criteria           | Marks |
|--------------------|-------|
| • Correct solution | 1     |

**Sample answer**

$$\ddot{x} = 6(t - 1)$$

this is a linear function with a positive gradient

$\therefore$  minimum acceleration when  $t = 0$ .

$$\begin{aligned}
 \therefore \ddot{x} &= 6(0 - 1) \\
 &= -6 \text{ cm s}^{-2}
 \end{aligned}$$

12 (b) (v)

| Criteria                                                                                                                                                                                                                                                    | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correct solution, including comment on location, velocity (stationary point) and acceleration.</li> </ul>                                                                                                            | 2     |
| <ul style="list-style-type: none"> <li>Partially correct solution including comment on <b>two</b> of the following: <ul style="list-style-type: none"> <li>- Location</li> <li>- Velocity (stationary point)</li> <li>- Acceleration</li> </ul> </li> </ul> | 1     |

**Sample answer**

The particle starts 1cm to the left of the origin. It slows down as it moves towards the origin and stops once it reaches it. It then accelerates as it moves the right, away from the origin.

12 (c)

| Criteria                                                           | Marks |
|--------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correct solution</li> </ul> | 1     |

**Sample answer**

$$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1 - t^2}{1 + t^2} = \cos 2\theta \quad (\text{when } t = \tan \theta)$$

12 (d)

| Criteria                                                           | Marks |
|--------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correct solution</li> </ul> | 1     |

**Sample answer**

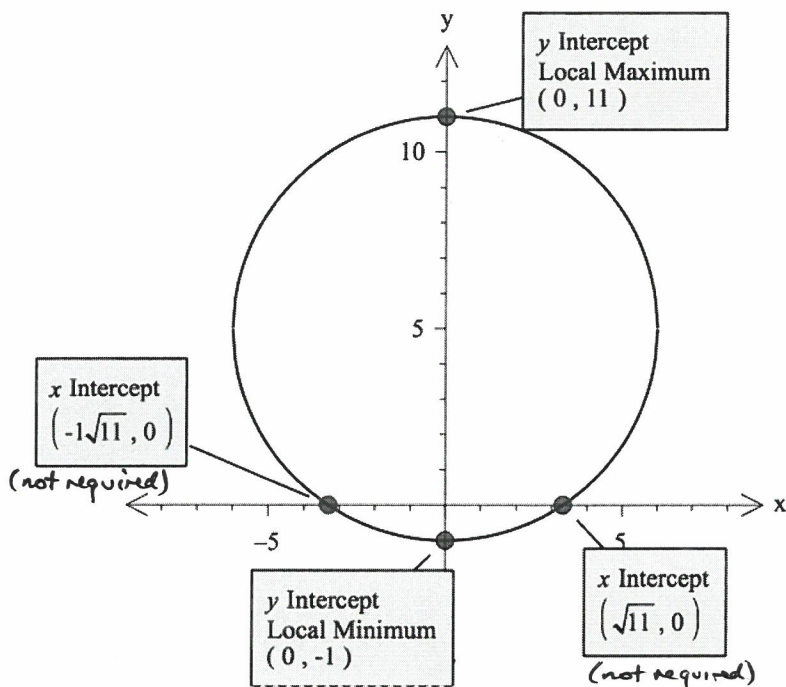
All real  $x$ ,

ie  $(-\infty, \infty)$

12 (e) (i)

| Criteria                                                               | Marks |
|------------------------------------------------------------------------|-------|
| • Correct graph, labelling centre and <u>y intercept</u>               | 2     |
| • Correct equation: $x^2 + (y-5)^2 = 36$ , after completing the square | 1     |

Sample answer



12 (e) (ii)

| Criteria                         | Marks |
|----------------------------------|-------|
| • Correct solution               | 2     |
| • Some progress towards solution | 1     |

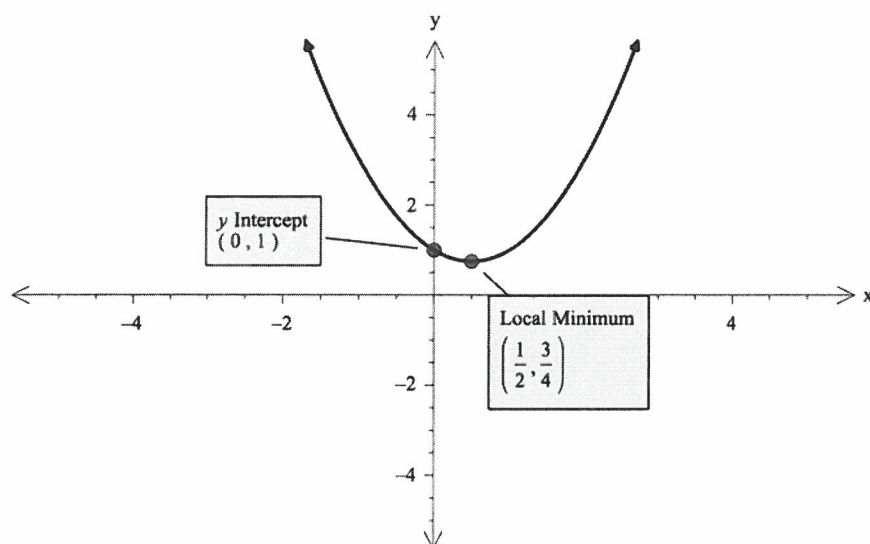
Sample answer

$$\begin{aligned}
 (6\sin\theta)^2 + (y-5)^2 &= 36 \\
 36\sin^2\theta + (y-5)^2 &= 36 && | \\
 (y-5)^2 &= 36(1 - \sin^2\theta) \\
 (y-5)^2 &= 36\cos^2\theta \\
 y-5 &= \pm 6\cos\theta \\
 y &= 5 \pm 6\cos\theta && |
 \end{aligned}$$

13 (a) (i)

| Criteria                                             | Marks |
|------------------------------------------------------|-------|
| • Correct graph (critical point labels not required) | 2     |
| • Correct shape (concave up parabola)                | 1     |

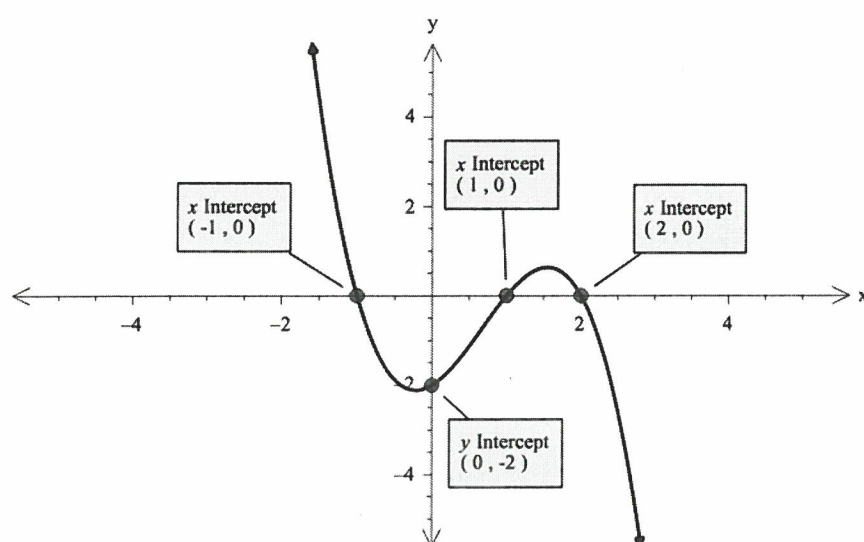
Sample answer



13 (a) (ii)

| Criteria                                                                                   | Marks |
|--------------------------------------------------------------------------------------------|-------|
| • Correct graph with intercepts shown in correct positions (intercept values not required) | 2     |
| • Correct shape (negative cubic)                                                           | 1     |

Sample answer



13 (b) (i)

| Criteria                                                           | Marks |
|--------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correct solution</li> </ul> | 1     |

**Sample answer**

$$\begin{aligned}
 \text{LHS} &= \cos(A + B) + \cos(A - B) \\
 &= \cos A \cos B - \sin A \sin B + \cos A \cos B + \sin A \sin B \\
 &= 2 \cos A \cos B
 \end{aligned}$$

13 (b) (ii)

| Criteria                                                                                                                                                   | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Correct solution</li> </ul>                                                                                         | 3     |
| <ul style="list-style-type: none"> <li>Correct method with 1 error</li> </ul>                                                                              | 2     |
| <ul style="list-style-type: none"> <li>Uses part i) to rewrite expression as <math>= 2 \cdot \cos \frac{\pi}{4} \cdot \cos \frac{\pi}{6}</math></li> </ul> | 1     |

**Sample answer**

$$\begin{aligned}
 &\cos \frac{5\pi}{12} + \cos \frac{\pi}{12} \\
 &= \cos \left( \frac{\pi}{4} + \frac{\pi}{6} \right) + \cos \left( \frac{\pi}{4} - \frac{\pi}{6} \right) \\
 &= 2 \cdot \cos \frac{\pi}{4} \cdot \cos \frac{\pi}{6} \quad \text{using } i) \\
 &= 2 \cdot \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} \\
 &= \frac{\sqrt{3}}{\sqrt{2}} \\
 &= \frac{\sqrt{6}}{2}
 \end{aligned}$$

**Question 13** (continued)

**Marks**

(c) The number of bacteria in a culture is given by  $N = N_0 e^{0.28t}$  where  $t$  is in hours.

- (i) If there are initially 100 bacteria, how many will there be after 10 hours? 1

$$N = 100 e^{0.28(10)}$$

$$= 1644 \text{ (nearest whole)}$$

- (ii) How many hours will it take for the population to reach 200 000? Answer to the nearest hour. 2

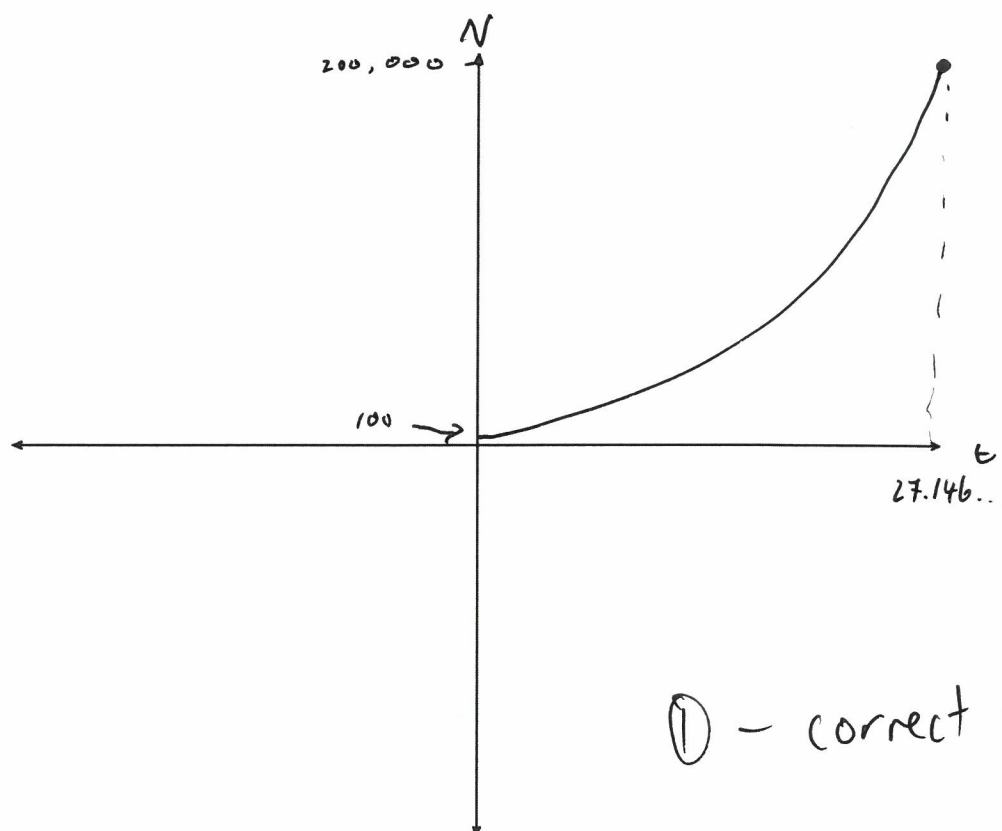
$$200000 = 100 e^{0.28t}$$

$$\frac{\ln 2000}{0.28} = t \quad \textcircled{1}$$

$$t = 27.146 \dots$$

$$= 28 \text{ h (as when } t = 27, N < 200,000 \text{)}. \quad \textcircled{1}$$

- (iii) Graph the relationship between  $N$  and  $t$  for  $N \in [100, 200\,000]$  1



**Question 13 continues on the next page**

13 (d)

| Criteria                                                                               | Marks |
|----------------------------------------------------------------------------------------|-------|
| • Correct solution, including explanation of what is happening.                        | 3     |
| • Correctly calculates the rate of change of the surface area when the radius is 100mm | 2     |
| • Correctly applies the chain rule                                                     | 1     |

**Sample answer**

$$\frac{dr}{dt} = -0.2$$

$$SA = 4\pi r^2$$

$$\frac{dSA}{dr} = 8\pi r$$

$$\therefore \frac{dSA}{dt} = \frac{dSA}{dr} \times \frac{dr}{dt}$$

$$= -0.2 \times 8\pi r$$

$$= -\frac{8\pi r}{5}$$

when  $r = 100$ ,

$$\frac{dSA}{dt} = -\frac{8\pi(100)}{5}$$

$$= -160\pi \text{ mms}^{-1}$$

$\therefore$  the surface area is decreasing by  $160\pi$  mm per second.

**Question 14**

14 (a)

| Criteria                         | Marks |
|----------------------------------|-------|
| • Correct solution               | 2     |
| • Some progress towards solution | 1     |

**Sample answer**

$$\text{let } y = \sin^{-1}(-x)$$

$$\therefore \sin y = -x, \text{ for } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$x = -\sin y$$

$$\therefore x = \sin(-y)$$

$$\therefore \sin^{-1}x = -y$$

$$y = -\sin^{-1}x$$

$$\therefore \sin^{-1}(-x) = -\sin^{-1}x$$

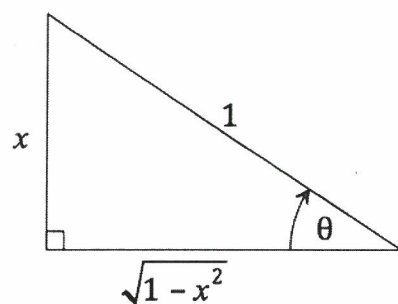
14 (b)

| Criteria                        | Marks |
|---------------------------------|-------|
| • Correct solution              | 2     |
| • Correctly constructs triangle | 1     |

**Sample answer**

let  $\theta = \sin^{-1} x$

$\therefore$  constructing triangle:



By Pythagoras' Thm, base =  $\sqrt{1^2 - x^2}$

$\therefore$  using trigonometric ratios,

$$\tan \theta = \frac{x}{\sqrt{1-x^2}}$$

**Question 14** (continued)

**Marks**

(c) Given  $P(x) = 6x^3 - 11x^2 - 3x + 2$

(i) Show  $x - 2$  is a factor of  $P(x)$

1

$$\begin{aligned} P(2) &= 6(2)^3 - 11(2)^2 - 3(2) + 2 \\ &= 48 - 44 - 6 + 2 \\ &= 0. \end{aligned}$$

$\therefore x - 2$  is a factor of  $P(x)$ .

(ii) Hence, factorise  $P(x)$  and find the roots of  $P(x) = 0$

3

$$\begin{array}{r} 6x^2 + 7x - 1 \\ x - 2 \overline{) 6x^3 - 11x^2 - 3x + 2} \\ \underline{6x^3 - 12x^2} \phantom{- 3x + 2} \\ 7x^2 - 3x + 2 \\ \underline{7x^2 - 14x} \phantom{+ 2} \\ 11x + 2 \\ \underline{11x - 22} \\ 24 \end{array}$$

$$\begin{aligned} \therefore P(x) &= (x - 2)(6x^2 + 7x - 1) \\ &= (x - 2)(3x - 1)(2x + 1) \end{aligned}$$

$\therefore$  for  $P(x) = 0$ .

$$x = -\frac{1}{2}, \frac{1}{3}, 2$$

**Question 14 continues on the next page**

**Question 14** (continued)

**Marks**

- (d) Consider the polynomial  $P(x) = (x - 2)(x - 5)Q(x) + ax + b$ , where  $Q(x)$  is a polynomial and  $a$  and  $b$  are real numbers.

When  $P(x)$  is divided by  $(x - 2)$  the remainder is 4, and when it is divided by  $(x - 5)$ , the remainder is 13.

Find  $a$  and  $b$

2

$$\begin{aligned} P(2) &= 4 & \therefore 2a + b &= 4 & \textcircled{1} \\ P(5) &= 13 & \therefore 5a + b &= 13 & \textcircled{2} \\ & & \textcircled{2} - \textcircled{1} & & \\ & & 3a &= 9 & \\ & & \therefore a &= 3 & \leftarrow \textcircled{1} \end{aligned}$$

**Question 14 continues on the next page**

$$\begin{aligned} \therefore 2(3) + b &= 4 \\ b &= -2 & \leftarrow \textcircled{1} \end{aligned}$$

14 (e) (i)

| Criteria                                   | Marks |
|--------------------------------------------|-------|
| • Correct solution                         | 2     |
| • Correctly applies simultaneous equations | 1     |

**Sample answer**

when  $t = 0$   $T = 31.1$

$$\therefore 31.1 = P + Ae^{k \cdot 0}$$

$$\therefore 31.1 = P + A \quad (1)$$

when  $t = 10$   $T = 30.7$

$$\therefore 30.7 = P + Ae^{10k} \quad (2)$$

$$(1) - (2)$$

$$0.4 = A - Ae^{10k}$$

$$Ae^{10k} = A - 0.4$$

$$e^{10k} = \frac{A - 0.4}{A}$$

$$10k = \ln\left(\frac{A - 0.4}{A}\right)$$

$$k = \frac{\ln\left(\frac{A - 0.4}{A}\right)}{10}$$

14 (e) (ii)

| Criteria           | Marks |
|--------------------|-------|
| • Correct solution | 1     |

**Sample answer**

when  $t = 0$   $T = 31.1$

$$\therefore 31.1 = 26 + Ae^{k \cdot 0}$$

$$\therefore A = 5.1$$

$\therefore$  using  $i)$

$$k = \frac{\ln\left(\frac{5.1 - 0.4}{5.1}\right)}{10}$$

$$k = -0.00816....$$

$$k = -0.008 \text{ (3 d.p.)}$$

14 (e) (iii)

| Criteria                   | Marks |
|----------------------------|-------|
| • Correct solution         | 2     |
| • Correctly calculates $t$ | 1     |

**Sample answer**

when  $T = 38.6$

$$38.6 = 26 + 5.1 \cdot e^{-0.008t}$$

$$12.6 = 5.1 \cdot e^{-0.008t}$$

$$\frac{12.6}{5.1} = e^{-0.008t}$$

$$\ln\left(\frac{12.6}{5.1}\right) = -0.008t$$

$$t = -\frac{\ln\left(\frac{12.6}{5.1}\right)}{0.008}$$

$$t = -113.057... \text{ minutes}$$

$\therefore$  time of death = 8 pm - 113.057... minutes  
= 6:07 pm